

Lösung der Bewegungsgleichung für die Erdbahn um die Sonne

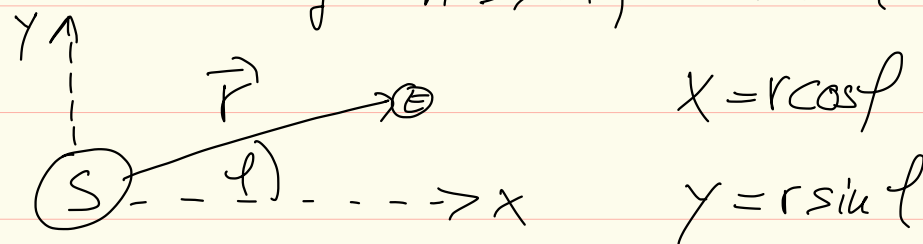
Skript Seite 165, Kapitel 7.2 Himmelsmechanik

2 Körper, keine äusseren Kräfte, keine äusseren Drehmomente

$\Rightarrow$ Impulserhaltung: $\vec{p}_{\text{tot}} = m \cdot \vec{v} + M \vec{v} = 0$

$\Rightarrow$ Drehimpulserhaltung: $\vec{L}_{\text{tot}} = \vec{r} \times \vec{p} = m(\vec{r} \times \vec{v}) = \text{konst.}$

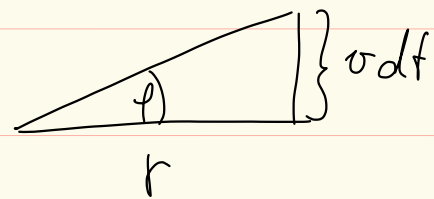
Näherung: $M \gg m$, $v \gg u$ (Sonne viel schwerer als Erde)



$$x = r \cos \phi$$

$$y = r \sin \phi$$

Geschwindigkeit: $v^2 = \dot{x}^2 + \dot{y}^2 = \frac{d}{dt} (r(t) \cdot \cos \phi(t))^2 + \frac{d}{dt} (r(t) \cdot \sin \phi(t))^2$
 $= (\dot{r} \cos \phi - r \dot{\phi} \sin \phi)^2 + (\dot{r} \sin \phi + r \dot{\phi} \cos \phi)^2$
 $= \dot{r}^2 + r^2 \dot{\phi}^2$



überstrichene Fläche: $dA = \frac{1}{2} |\vec{r} \times \vec{v}| dt$

$$= \left| \frac{\vec{L}_{\text{tot}}}{2m} \right| dt$$

$$= \frac{1}{2} r \cdot v \cdot dt$$

$$= \frac{1}{2} r \cdot r \frac{d\phi}{dt} dt$$

$$= \frac{1}{2} r^2 \dot{\phi} dt$$

$$\Rightarrow \dot{A} = \frac{dA}{dt} = \frac{L_{\text{tot}}}{2m} = \text{konst.}$$

$$= \frac{1}{2} r^2 \dot{\phi}$$

$$\Rightarrow \dot{\phi} = L / m r^2$$

Energieerhaltung: $E = \frac{1}{2} m v^2 - G_N \frac{mM}{r} = \text{konst.}$

$$= \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\phi}^2) - G_N \frac{mM}{r}$$

$$= \frac{1}{2} m \left(\dot{r}^2 + \frac{L^2}{m^2 r^2} \right) - G_N \frac{mM}{r}$$

Neue Erhaltungsgrößen (Abkürzungen):

$$a = -G_N \frac{mM}{2E}$$

$$\varepsilon^2 = 1 + \frac{2EL^2}{G_N^2 M^2 m^3}$$

Energieerhaltung aufgelöst nach $\dot{r}^2$:

$$\frac{1}{2} m \dot{r}^2 = E + G_N \frac{mM}{r} - \frac{1}{2} m \frac{L^2}{m^2 r^2}$$

$$\Rightarrow \dot{r}^2 = \underbrace{\frac{2E}{m}}_{= -\frac{G_N M}{a}} + 2G_N \frac{M}{r} - \frac{L^2}{m^2 r^2}$$

$$\Rightarrow \dot{r}^2 = \frac{G_N M}{a} \left(-1 + 2 \frac{a}{r} - \frac{L^2}{m^2 r^2} \frac{a}{G_N M} \right)$$

$$= \frac{G_N M}{a} \frac{1}{r^2} \left(-r^2 + 2ar - \frac{L^2 a}{m^2 G_N M} \right)$$

A

Berechne: $a^2 \varepsilon^2 - (r-a)^2 = G_N^2 \frac{m^2 M^2}{4E^2} \left(1 + \frac{2EL^2}{G_N^2 M^2 m^3}\right) - \left(r + G_N \frac{mM}{2E}\right)^2$

$$= G_N^2 \frac{m^2 M^2}{4E^2} \frac{2EL^2}{G_N^2 M^2 m^3} - 2r G_N \underbrace{\frac{mM}{2E}}_{-a} - r^2$$

$$= -r^2 + 2ar + \underbrace{\frac{1}{m} \frac{1}{2} \frac{1}{E} L^2}_{-L^2 \frac{a}{G_N m^2 M}} = A \text{ (vergleiche mit vorher)}$$

$$\Rightarrow \dot{r} = \frac{dr}{dt} = \sqrt{\frac{G_N M}{a}} \frac{1}{r} \sqrt{a^2 \varepsilon^2 - (r-a)^2}$$

Lösung der Differentialgleichung durch Separation:

$$\int \frac{r dr}{\sqrt{a^2 \varepsilon^2 - (r-a)^2}} = \sqrt{\frac{G_N M}{a}} \int dt$$

Substitution: $r(t) = a(1 - \varepsilon \cos \alpha(t))$ (neue Variable α)
 $\Rightarrow dr = a \varepsilon \sin \alpha d\alpha$

Berechne in $\alpha(t)$:

$$\begin{aligned} a^2 \varepsilon^2 - (r-a)^2 &= a^2 \varepsilon^2 - [a(1 - \varepsilon \cos \alpha) - a]^2 \\ &= a^2 \varepsilon^2 - a^2 [1 - \varepsilon \cos \alpha - 1]^2 \\ &= a^2 \varepsilon^2 - a^2 \varepsilon^2 \cos^2 \alpha \\ &= a^2 \varepsilon^2 \sin^2 \alpha \end{aligned}$$

$$\int \frac{r dr}{\sqrt{a^2 \varepsilon^2 - (r-a)^2}} = \int \frac{a(1 - \varepsilon \cos \alpha) \cdot a \cdot \varepsilon \sin \alpha d\alpha}{\sqrt{a^2 \varepsilon^2 \sin^2 \alpha}}$$

$$\begin{aligned} &= \int a(1 - \varepsilon \cos \alpha) d\alpha \\ &= a(\alpha - \varepsilon \sin \alpha) + \text{konst} \\ &= \sqrt{\frac{G_N M}{a}} \cdot t = \sqrt{\frac{G_N M}{a^3}} a t \end{aligned}$$

$$\alpha - \varepsilon \sin \alpha = \sqrt{\frac{G_N M}{a^3}} \cdot t + \text{konst}$$

Ziel: Löse Differentialgleichung für $\alpha(t)$

Ableitung nach t : $\frac{d\alpha}{dt} - \varepsilon \cos \alpha \frac{d\alpha}{dt} = \sqrt{\frac{G_N M}{a^3}}$

$$\Rightarrow \frac{d\alpha}{dt} = \frac{1}{1 - \varepsilon \cos \alpha} \sqrt{\frac{G_N M}{a^3}}$$

Berechnung von $\dot{\varphi}$: $\dot{\varphi} = \frac{d\varphi}{dt} = \frac{L}{mr^2} = \frac{L}{ma^2(1-\varepsilon \cos \alpha)^2}$

Substitution: $\frac{d\varphi}{dt} = \frac{d\varphi}{d\alpha} \frac{d\alpha}{dt} = \frac{d\varphi}{d\alpha} \sqrt{\frac{GM}{a^3}} \frac{1}{1-\varepsilon \cos \alpha}$

$$= \frac{L}{ma^2(1-\varepsilon \cos \alpha)^2}$$

=> Ausdruck für $\frac{d\varphi}{d\alpha}$: $\frac{d\varphi}{d\alpha} = \frac{L}{ma^2} \sqrt{\frac{a^3}{GM}} \frac{1}{(1-\varepsilon \cos \alpha)}$

Berechne: $\sqrt{1-\varepsilon^2} = \sqrt{-\frac{2EL^2}{G_N^2 M^2 m^3}} = \sqrt{-\frac{2E}{mMG_N} \cdot \frac{L^2}{G_N M m^2}}$

$$= \sqrt{\frac{L^2}{G_N M m^2 a}} = \frac{L}{ma^2} \sqrt{\frac{a^3}{GM}}$$

=> $\frac{d\varphi}{d\alpha} = \frac{\sqrt{1-\varepsilon^2}}{1-\varepsilon \cos \alpha}$

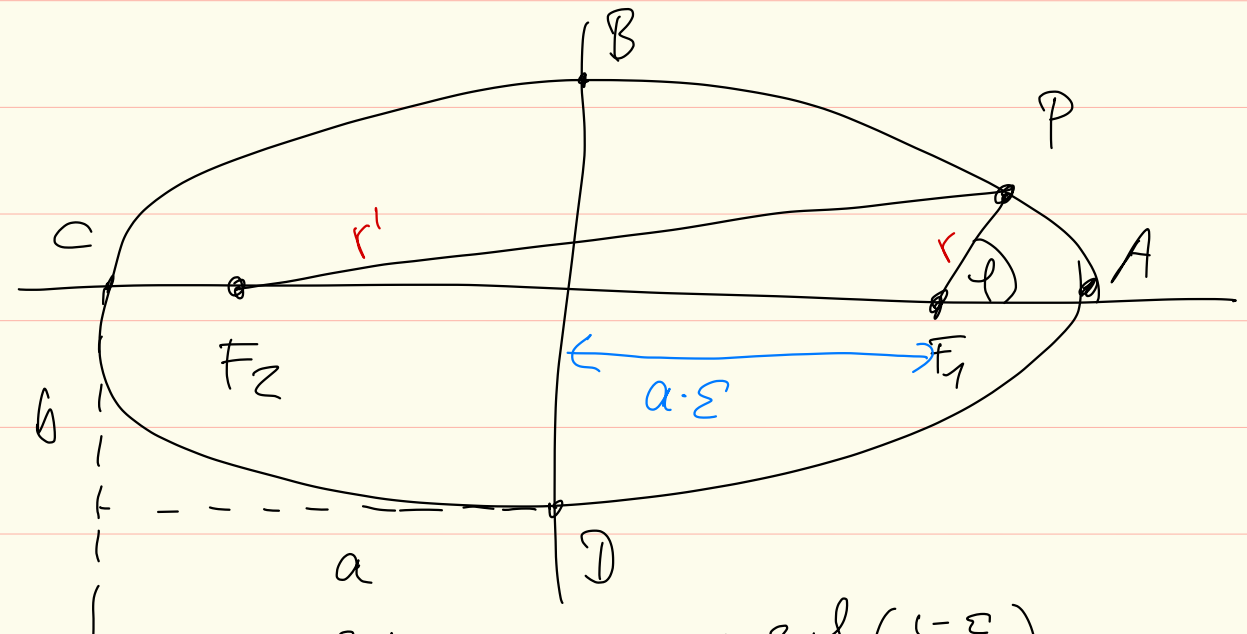
Separation der Variablen: $\int d\varphi = \sqrt{1-\varepsilon^2} \int \frac{d\alpha}{1-\varepsilon \cos \alpha}$

$$\frac{2}{\sqrt{1-\varepsilon^2}} \arctan \left[\frac{(1+\varepsilon) \tan \frac{\alpha}{2}}{\sqrt{1-\varepsilon^2}} \right]$$

$$\frac{\varphi}{2} = \arctan \left[\sqrt{\frac{1+\varepsilon}{1-\varepsilon}} \tan \frac{\alpha}{2} \right]$$

=> $\tan \frac{\varphi}{2} = \sqrt{\frac{1+\varepsilon}{1-\varepsilon}} \tan \frac{\alpha}{2}$

Kreis: $\varepsilon = 0 \Rightarrow \varphi = \alpha$



$$\cos \alpha = \frac{1 - \tan^2 \frac{\alpha}{2}}{1 + \tan^2 \frac{\alpha}{2}} = \frac{1 - \tan^2 \frac{\varphi}{2} \left(\frac{1-\varepsilon}{1+\varepsilon} \right)}{1 + \tan^2 \frac{\varphi}{2} \left(\frac{1-\varepsilon}{1+\varepsilon} \right)}$$

$$= 1 - \frac{1 - \cos \varphi}{1 + \cos \varphi} \left(\frac{1-\varepsilon}{1+\varepsilon} \right) = \frac{(1 + \cos \varphi)(1 + \varepsilon) - (1 - \cos \varphi)(1 - \varepsilon)}{(1 + \cos \varphi)(1 + \varepsilon) + (1 - \cos \varphi)(1 - \varepsilon)}$$

$$= \frac{\varepsilon + \cos \varphi}{1 + \varepsilon \cos \varphi}$$